Data Science

Lecture 06: Decision Trees and Ensembles

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Where We Left Off

Lectures 04–05 gave us models that predict and cannot explain themselves — a few hundred weights, or millions. Today: a model you can read aloud, and what it costs to keep it readable.

Lecture 05
accurate, opaque

→

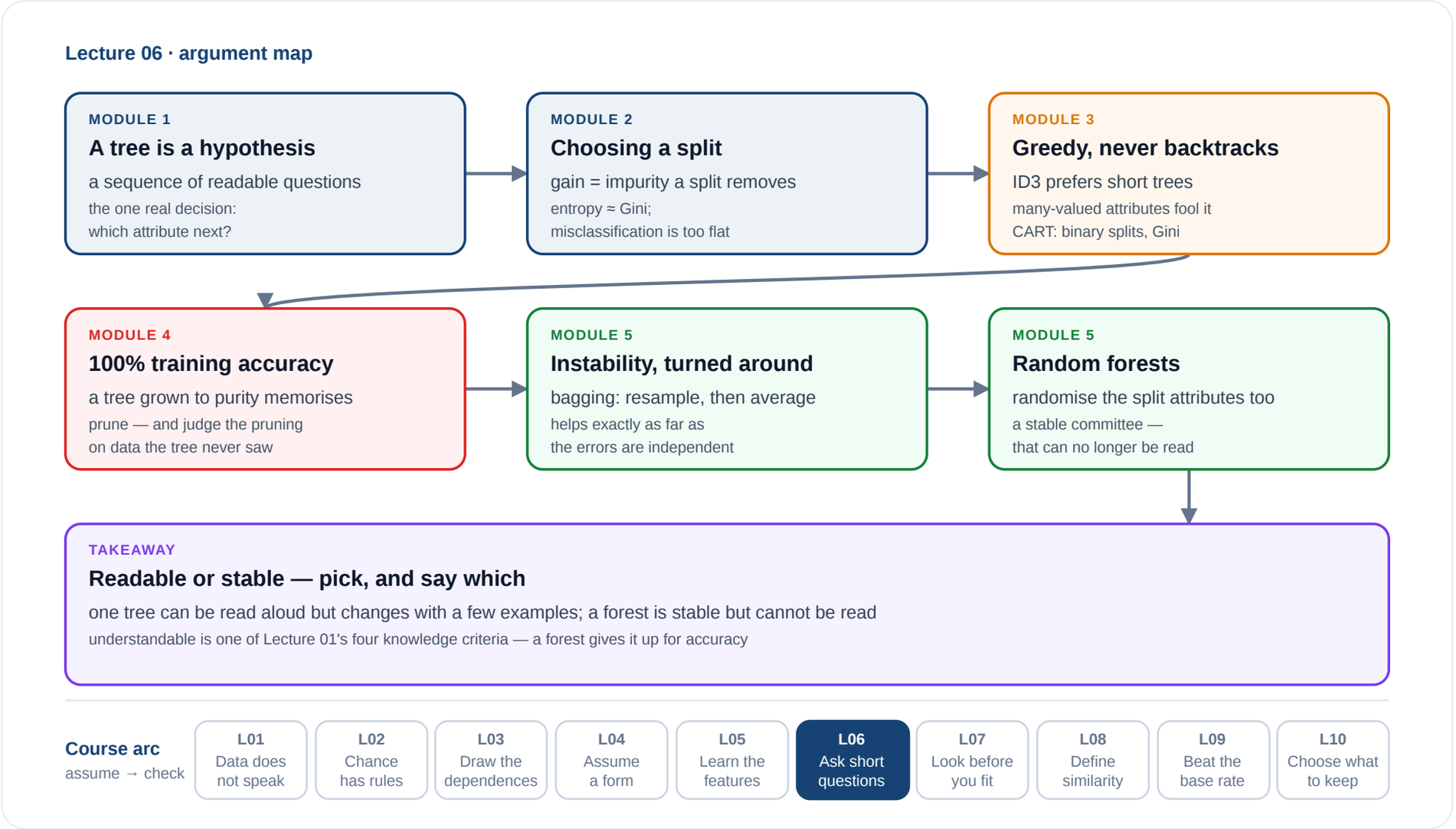
Modules 1–4
one tree: readable, unstable

→

Module 5
a forest: stable, opaque again

> "Ultimately understandable" was one of Lecture 01's four criteria for knowledge. This lecture is the first to trade it away on purpose.

Argument Map



Module 1

Trees as Hypotheses

> ***Guiding Question:*** *A linear model hides its reasoning in 200 weights. What would a model look like that can be read aloud?*

A Tree Is a Sequence of Questions

Each inner node tests one attribute, each branch is an answer, each leaf is a class.

Example	Sky	Temperature	Humidity	Wind	EnjoySport
1	sunny	warm	normal	strong	yes
2	sunny	warm	high	strong	yes
3	rainy	cold	high	strong	**no**
4	sunny	warm	high	strong	yes

To classify a new day, start at the root and follow the answers to a leaf.

> The tree **is** the hypothesis. Unlike a weight vector, it can be read aloud as a set of rules — which is why trees are the standard choice when a decision has to be explained.

Construction and Classification

Construction *(recursive)*

If the data is pure enough, make a leaf. Otherwise: choose a **split criterion**, decompose D into $D_1, \ldots, D_s$, and recurse on each part.

Classification *(recursive)*

If the node is a leaf, return its class. Otherwise: follow the successor selected by the example's attribute value.

The whole learning problem reduces to one question: **which attribute to split on?** Everything else is bookkeeping.

When to Use Them

Well suited

Attribute-value data; discrete or continuous features; missing values; a need for an explainable model; no assumption of linearity.

Poorly suited

Smooth relationships (a tree approximates a diagonal boundary with a staircase); many equally relevant features; and — as we will see — stability.

Mini-Exercise

Find a tree with a **single test** that classifies the four EnjoySport examples from the first slide correctly. Is it the only one?



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> `Sky` works (sunny → yes, rainy → no) — and so does `Temperature` (warm → yes, cold → no). Both separate the data perfectly; even the next module's split criterion scores them equally (0.811 bits each). The data cannot choose: the learner's **inductive bias**, not the examples, breaks the tie.

Takeaway

A decision tree is a **readable hypothesis** built by recursive splitting. The only real decision is which attribute to split on next — and the data alone does not settle it.

Module 2

Choosing a Split

> ***Guiding Question:*** *Every attribute could be split on. What makes one split better than another — and how do we measure it?*

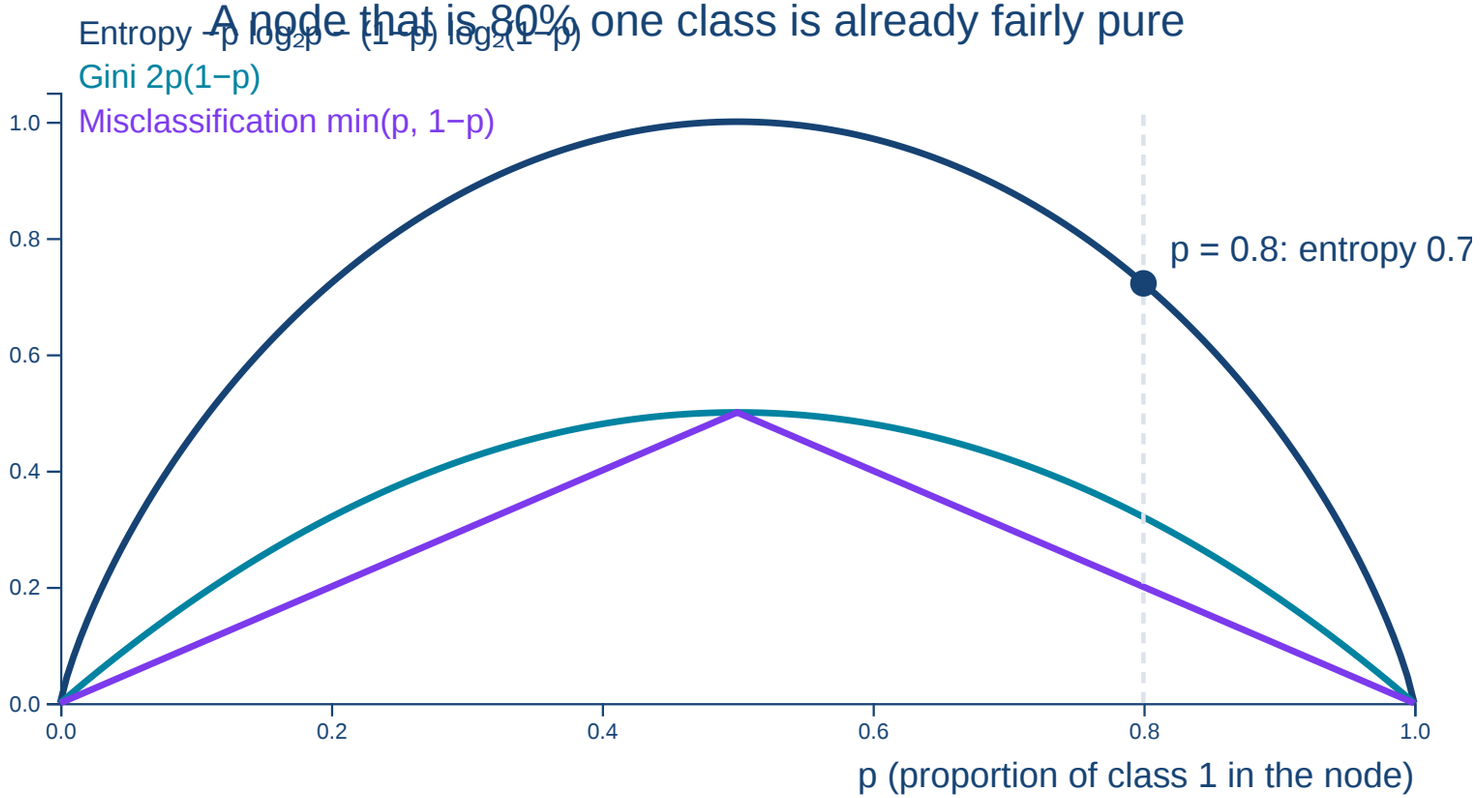
What Makes a Split Good?

A split is good if the resulting subsets are **purer** than what we started with — closer to containing a single class.

So we need a number that measures impurity: maximal when the classes are evenly mixed, zero when a node is pure.

Three Measures

A node that is 80% one class is already fairly pure



Entropy and Gini

For a node with class probabilities $P(A_1), \ldots, P(A_k)$:

$$H(\mathcal{A}) = -\sum_{i=1}^{k} P(A_i) \log_2 P(A_i) \qquad G(\mathcal{A}) = 1 - \sum_{i=1}^{k} P(A_i)^2$$

Entropy H is measured in bits — the average number of yes/no questions needed to identify an example's class. **Gini impurity** G is the probability that two examples drawn at random from the node belong to different classes.

	Pure node	Two classes, 50/50	k equal classes
Entropy H	0	1 bit	$\log_2 k$ bits
Gini G	0	0.5	$1 - 1/k$

Conditional Entropy and Information Gain

After splitting on an attribute $\mathcal{B}$ with outcomes $B_1, \ldots, B_s$:

$$H(\mathcal{A} \mid \mathcal{B}) = \sum_{j=1}^{s} P(B_j) \cdot H(\mathcal{A} \mid B_j)$$

— the impurity of each part, weighted by how much data lands in it.

$$\text{Gain}(\mathcal{B}) = H(\mathcal{A}) - H(\mathcal{A} \mid \mathcal{B})$$

Information gain is the impurity removed by the split. Choose the attribute that maximises it.

Worked Example: The Root of PlayTennis

Quinlan's 14 weather days: 9 yes, 5 no, so $H = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.940$ bits. Counts are (yes, no) per value.

Attribute	Split	$H(\mathcal{A} \mid \mathcal{B})$	Gain
Outlook	sunny (2, 3) · overcast (4, 0) · rain (3, 2)	0.694	**0.247**
Humidity	high (3, 4) · normal (6, 1)	0.788	0.152
Wind	weak (6, 2) · strong (3, 3)	0.892	0.048
Temperature	hot (2, 2) · mild (4, 2) · cool (3, 1)	0.911	0.029

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> Outlook wins the root although two of its branches are *more* mixed than the root (0.971 vs 0.940 bits): its overcast branch is **pure**, so 4 of the 14 days need no further question.

Mini-Exercise

A node holds 4 yes and 4 no. A split yields [3 yes, 1 no] and [1 yes, 3 no]; a 3:1 node has entropy 0.811. What is the information gain?



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> Before: $H = 1$ bit. After: both halves hold half the data and have $H = 0.811$, so $H(\mathcal{A} \mid \mathcal{B}) = 0.811$. $\mathrm{Gain} = 1 - 0.811 = 0.189$ bits — a split that "looks decent" removes less than a fifth of a bit.

Takeaway

Impurity measures how mixed a node is; **information gain is the impurity a split removes**, weighted by the data it moves. Entropy and Gini agree closely; misclassification error is too flat to choose between splits.

Module 3

Building the Tree

> ***Guiding Question:*** *ID3 never reconsiders a split. What does that greedy choice assume about the world?*

ID3

ID3(examples D, attributes A, target)

1. If all examples share one class $\rightarrow$ return a leaf with that class.
2. If A is empty $\rightarrow$ return a leaf with the majority class.
3. Otherwise choose $A^* \in A$ with the highest **information gain**.
4. For each value a of A^*: if D_a is empty, add a leaf with the majority class of D; otherwise add the subtree **ID3**(D_a, $A \setminus \{A^*\}$, target).

Greedy and recursive: each split is chosen for immediate gain, and never reconsidered.

What ID3 Assumes

Hypothesis space

All decision trees over the attributes — so the target function is certainly in there. No representational bias.

Search bias

It prefers trees that place **high-gain attributes near the root**, and therefore **shorter trees**. It stops at the first tree that fits.

Two assumptions, not facts. **Short trees generalise** — Occam's razor as an algorithm. And **a useful attribute shows gain on its own**: for $y = x_1 \operatorname{XOR} x_2$ on balanced data, each attribute alone has gain 0, yet together they decide y — greedy search cannot see it.

CART

What differs

Binary splits only; **Gini** impurity instead of entropy; handles **regression** by splitting on variance reduction and predicting the mean in each leaf.

What does not

The template is unchanged: choose a split, decompose, recurse. CART is a different set of choices inside the same skeleton.

Binary splits reduce a defect of multi-way splits: an attribute with many values (an ID number, in the extreme) achieves high information gain by fragmenting the data into pure singletons. C4.5 corrects for it with the **gain ratio**, the gain divided by the entropy of the split itself.

Mini-Exercise

Your dataset has a `customer_id` column, unique for every row. What information gain does a split on it achieve — and what does the resulting tree predict for a new customer?

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> The **maximum possible**: every branch holds one example, every leaf is pure, so the gain equals the root's entropy. And for a new customer — nothing: their id matches no branch. Perfect purity, zero generalisation; hence gain ratio, or binary splits.

Takeaway

ID3 is a **greedy, non-backtracking search** whose bias is a preference for short trees. CART changes the split type and the impurity measure, not the idea. Both can be fooled by an attribute with too many values.

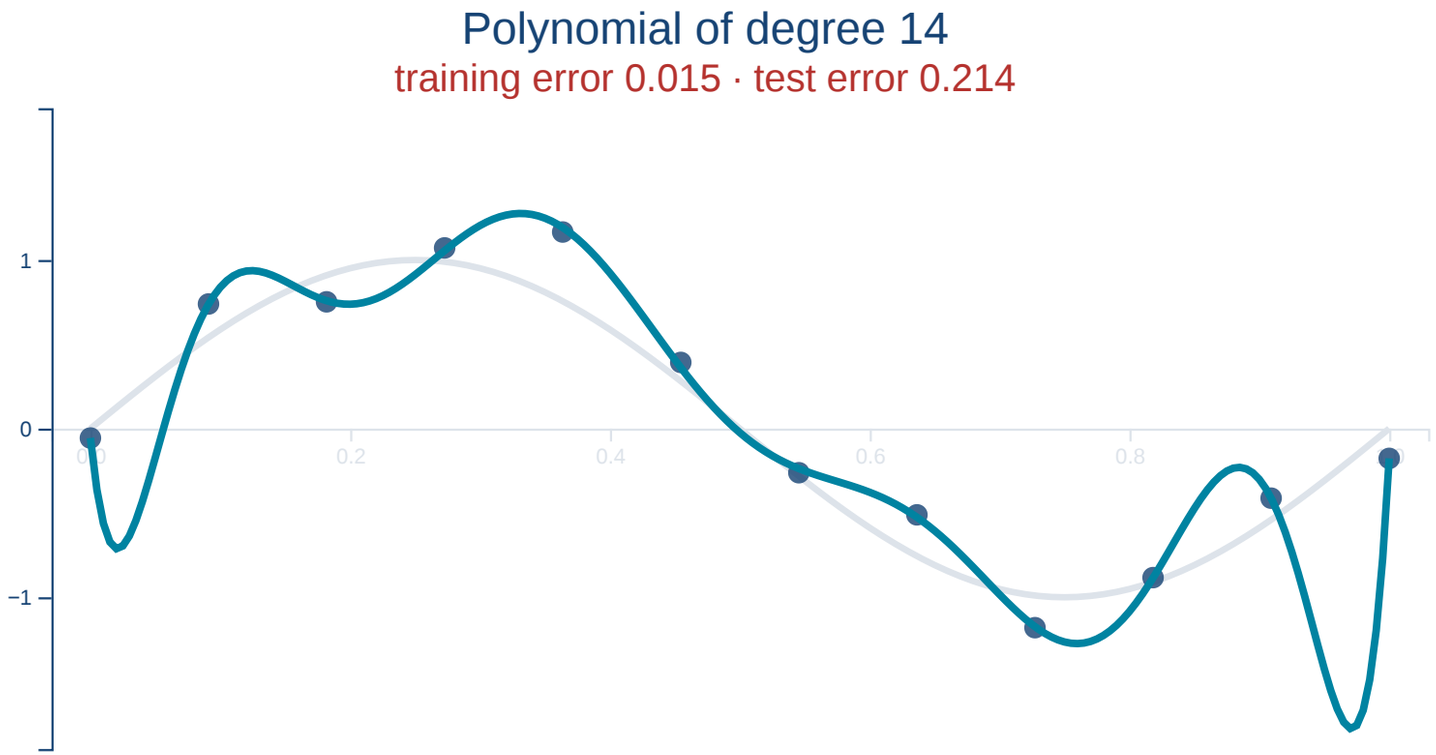
Module 4

Overfitting and Pruning

> ***Guiding Question:*** *A tree can reach 100% training accuracy on any dataset without contradictory examples. Why is that a symptom rather than an achievement?*

A Tree Grown to Purity Is Overfitted

Left alone, ID3 splits until every leaf is pure — memorising the training set, noise included.



Tree depth plays the role of polynomial degree: more splits, lower training error, and eventually rising test error.

Two Places to Intervene

Stopping (pre-pruning)

Halt splitting **during** construction — when a node is small, when gain is below a threshold, or at a depth limit.

Cheap, but short-sighted: a weak split may enable a strong one beneath it.

Pruning (post-pruning)

Grow the full tree, then **remove** subtrees that do not earn their keep.

More expensive, and generally more reliable.

Reduced Error Pruning

1. Split the data into a **training** and a **validation** set.
2. Grow the full tree on the training set.
3. For each inner node: consider replacing its subtree with a leaf carrying the majority class.
4. Keep the replacement if accuracy on the **validation** set does not fall.
5. Repeat until no beneficial replacement remains.

The decision is made on data the tree was **not** grown on. Pruning against the training set would remove nothing — every subtree helps there by construction.

Mini-Exercise

A tree with 340 leaves scores 100% on training data, 78% on validation. Pruned to 41 leaves it scores 94% and 89%. Training accuracy fell by six points — did pruning fail?

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> No. The 299 removed leaves fitted noise — examples that are now misclassified, as intended. Unseen data gained **11 points**, and the tree became small enough to read. Training accuracy was never the objective.

Takeaway

> An unpruned tree memorises. **Pruning trades training accuracy for generalisation**, and the trade must be judged on data the tree never saw.

Module 5

Ensembles

> ***Guiding Question:*** *Change a few examples and the tree changes completely. Can that instability be turned into an advantage?*

Why One Learner Falls Short

Three reasons, after Dietterich (2000):

Statistical

Too little data for the hypothesis space: many hypotheses fit equally well, and small changes in the data pick a very different one.

Computational

The learner cannot search its hypothesis space exhaustively and settles for a heuristic path — a local optimum.

Representational

The model family cannot express the target at all; a combination of models may.

Decision trees are the textbook unstable learner (the statistical reason): change a handful of examples and a different attribute wins at the root, producing a completely different tree.

The Idea

Ensemble learning builds several learners and combines their predictions — by majority vote for classification, by averaging for regression.

Bagging

Bootstrap **agg**regat**ing** — train on resampled datasets.

Boosting

Train weak learners in sequence, each focusing on the previous one's mistakes. *Named here; not developed in this course.*

Random forest

Bagging plus randomised attribute selection at every split.

Ensembles exploit instability: **many different models, each better than chance and wrong in partly independent ways, are right together more often than any one of them**.

Bootstrap Aggregating

The problem. A group of predictors helps only if they differ — but we have one algorithm and one dataset, which yield one model.

The solution. Manufacture different datasets: draw n examples from D **with replacement**, train a predictor on each sample, and combine them by vote or average.

> The learner never changes. Only the data it sees does.

What Resampling Actually Does

Each bootstrap sample contains about **63.2%** of the distinct examples of D. The probability that a particular example is *never* drawn in n tries is

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow[n\to\infty]{} e^{-1} \approx 0.368$$

The gain

Each model sees a different two-thirds of the data, so their errors are partly independent. **Averaging reduces variance** — and therefore overfitting.

The left-out third

The ~36.8% not drawn is the *out-of-bag* sample, usable as a free validation set for that model.

Mini-Exercise

You bag 100 trees, each 70% accurate, and take a majority vote. What accuracy does the vote reach if all 100 trees are identical — and if their errors are independent?

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> Identical: **70%** — a hundred copies vote unanimously. Independent: the vote is wrong only if 50 or more trees err at once, which happens with probability 2.2×10^{-5} — about **99.998%**. Both conditions matter: each tree better than chance (> 50%), and errors that are not shared.

Takeaway

> Bagging manufactures diversity by **resampling the data**, and averages away variance. It helps exactly to the extent that the models' errors are independent — and not at all when they are not.

Why Bagging Trees Is Not Enough

Trees stay similar

If one attribute has much higher gain than the rest, it wins at the root of **every** bootstrap sample. The trees correlate, and averaging them buys little.

And it costs

Interpretability is gone — you can no longer read the model as a set of rules — while the computation of 100 near-identical trees is largely wasted.

The Fix: Randomise the Split Too

Random forest (Breiman 2001)

For each of T trees:

1. Draw a bootstrap sample from D.
2. Grow a tree — but **at every node**, choose the split from a random subset $A' \subset A$ of m_A attributes rather than from all of them.
3. Do not prune.

Combine by majority vote (classification) or average (regression).

The dominant attribute is simply **absent** from most nodes' candidate sets, so other attributes get to define the tree. Diversity is engineered rather than hoped for.

What You Get and What You Give Up

Gained

Strong accuracy with almost no tuning; out-of-bag error for free; a usable attribute-importance measure.

Lost

The readable single tree. A forest is a **committee**, and no member speaks for it.

Attribute importance from a forest is not a causal statement. It reports which attributes the trees found useful for splitting — under this data, this sampling, this correlation structure.

Mini-Exercise

A pruned tree reaches 86% accuracy, a random forest 91%. Which do you deploy for an internal churn model — and which for automated loan refusals that must be justified to applicants?

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> Churn: the **forest** — five points matter and nobody needs the model explained. Loans: the **tree**, or a forest inside a documented, auditable process; a refusal that cannot be explained may be neither acceptable nor lawful. The best model fits the **decision**, not the leaderboard.

Takeaway

Random forests **engineer diversity** by randomising the attributes available at each split, turning unstable trees into a stable committee — at the cost of the one thing a single tree was best at: being readable.

The Churn Shop: A Tree You Can Read Aloud

ONE TREE

→ Contract **monthly**?

→ No **app login** for 30 days?

→ Then: **likely to churn** — call them.

The retention team can check every step against what they know.

A FOREST OF 500 TREES

Usually more accurate, and stable when a few customers change.

Its answer is a vote. Asked *why* this customer, it has no sentence to offer.

The same data, two answers to Lecture 01's criteria: the tree is **understandable** and fragile, the forest is accurate and unreadable. Say which one you chose, and why.